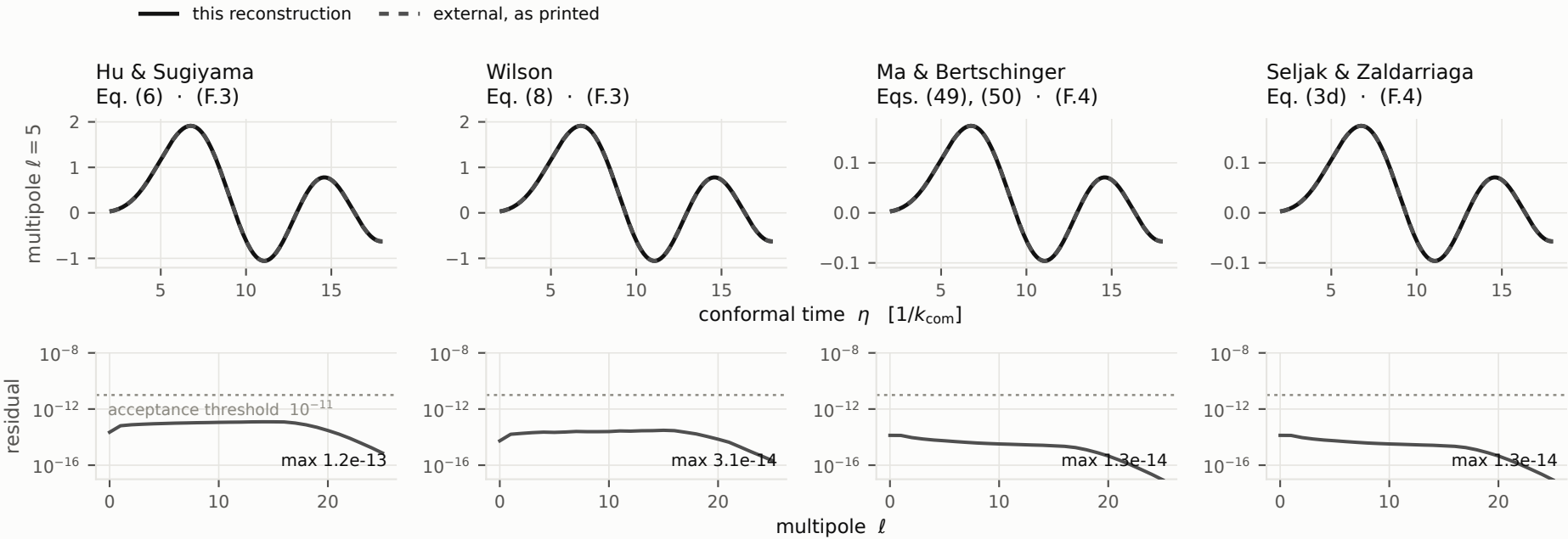
F1 The covariant free-streaming recursion against four external hierarchies



Each external hierarchy is transcribed from its own paper, not through Annals II Appendix F. Hu & Sugiyama and Wilson write the recursion in the β normalisation, Ma & Bertschinger and Seljak & Zaldarriaga in the α normalisation — which is why the upper panels do not share a vertical scale. Wilson writes it in the imaginary convention, with both bracket terms positive under an overall $-i$; $\Theta_\ell = i^\ell \delta_\ell$ recovers the real, opposite-sign form. That panel therefore tests the phase convention and not only the ℓ-weights. Overall scale is exactly 1; nothing is fitted.